

A dissertation on
**MEMORY-AWARE AUTOMATIC
DIFFERENTIATION FOR
OPTIMIZATION-BASED IMAGE
SUPER-RESOLUTION**

submitted in partial fulfillment for the award of the degree of

Master of Science in Data Science

by

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I hereby declare that the dissertation - II entitled **Memory-Aware Automatic Differentiation for Optimization-Based Image Super-Resolution** submitted by me to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfillment of the requirements of the award of the degree of **Master of Science in Data Science** is a bona fide record of the work carried out by me under the supervision of **Dr. Kaliyappan M.**

I further declare that the work reported in this dissertation-II, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this Institute or of any other Institute or University.

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CERTIFICATE

This is to certify that the dissertation-II titled “**Memory-Aware Automatic Differentiation for Optimization-Based Image Super-Resolution**” is prepared and submitted by **Chakka Ganesh Venkat Raghava (Reg. No. 24MDT1080)** to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Master of Science in Data Science** is a bona fide record carried out by him under my supervision, as per the VIT code of academics and research ethics.

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ABSTRACT

Image Super-Resolution is a term that is used to refer to the process of converting low-resolution images into high-resolution images, which have more definition and hard lines and all visual details. It plays a very important role in medical imaging and surveillance systems as well as satellite imaging and mobile photography usage in the world. The existing super-resolution algorithms based on deep learning require multi-million computational resources and huge memory, which makes implementation in scenarios that require real-time processing and scarce resources impossible. The project is offering a memory efficient image super-resolution model with a combination of lightweight deep learning model along with optimization based refinement algorithms. The basic model is Fast Super-Resolution Convolutional Neural Network (FSRCNN) since it does not consume a lot of computing resources but provides a high processing speed. The FSRCNN scheme generates a high-resolution original image on the given low-resolution image input. The framework employs the Automatic Differentiation (AD) and Conjugate Gradient (CG) optimization tools to improve the quality of the generated image. Automatic Differentiation is very efficient in calculating the gradients whereas the system employs the method to produce refinement of outputs by minimizing loss functions which leads to lower noise levels and structural integrity. Conjugate Gradient method is an implementation based on the past gradient values to accelerate the process as it sharpens the features in the image. Deep learning with optimization techniques allow the system to generate a higher quality of images with a lower processing demand. The given approach does not only improve visual clarity, but also preserves the essential elements of images but consumes less memory space than the traditional approaches.

Keywords: Image Super-Resolution, FSRCNN, Automatic Differentiation (AD), Conjugate Gradient (CG), PSNR, SSIM.

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List of Symbols, Abbreviations and Nomenclature

SR	: Image Super-Resolution
LR	: Low-Resolution Image
HR	: High-Resolution Image
CNN	: Convolutional Neural Network
FSRCNN	: Fast Super-Resolution Convolutional Neural Network
SRCNN	: Super-Resolution Convolutional Neural Network
ESPCN	: Efficient Sub-Pixel Convolutional Neural Network
UNet	: U-shaped Convolutional Network
AD	: Automatic Differentiation
CG	: Conjugate Gradient
MSE	: Mean Squared Error
PSNR	: Peak Signal-to-Noise Ratio
SSIM	: Structural Similarity Index Measure
x	: Input Image
x_0	: Output Image from FSRCNN
x_{final}	: Final Enhanced Image
y_{pred}	: Predicted Output Image
y_{true}	: Ground Truth Image
θ	: Model Parameters
η	: Learning Rate
α	: Step Size
λ	: Regularization Parameter
g_k	: Gradient at iteration k
d_k	: Search Direction in CG
β_k	: CG Update Coefficient
$TV(x)$	: Total Variation Function
∇L	: Gradient of Loss Function

Chapter 1

Introduction

Image Super-Resolution (SR) is a challenging problem in image processing and computer vision, the goal of which is to obtain high-resolution (HR) image of a poor-quality (LR) input image. Under real-life conditions, poor quality sensors, compression and bandwidth planning a poor image quality is obtained to obtain a clear and detailed high-resolution output. In order to obtain clear and detailed high-resolution output it is necessary to manage to enhance such images. The applications of it in super resolution include medical imaging, surveillance systems, satellite imaging and mobile photography, where a clearer image can be used to improve the diagnosis of an object by a physician, whereas a better image can be used to improve the recognition of objects and faces by the surveillance system. Similarly, in satellite imaging, a resolution image is applied to enhance the mapping and monitoring of the environment. In the recent years, image super-resolution tasks have attempted to achieve a high level of improvement by deep learning-based mapping SRCNN, FSRCNN and ESPCN have been demonstrated to learn high-resolution to low-resolution and low-resolution to high-resolution mappings. However, the majority of these models are inefficient because they have extremely big memory and computational overheads that suggest that they cannot be easily deployed in resource-constrained systems such as mobile devices and embedded systems. To address this deficiency, this project will target to develop a memory efficient super-resolution architecture. SRCNN is the base model since it is built upon lightweight deep neural network, FSRCNN due to its relatively fast computation and fewer memory

requirements. The suggested model is not a complication to the model but maximizes the output with the refinement techniques. Specifically, the calculation of the gradients is performed with the help of the so-called Automatic Differentiation (AD) that is employed to refine the generated image better and introduce the consistency and reduce noise. The Conjugate Gradient (CG) method is also used as an optimization method to enhance further convergence and stability in the process of refining. It is a configuration of all these optimization methods with a lean deep learning network that makes the system achieve a higher image quality at a fractional cost of the computational one. The success of the suggested method is determined using the traditional measures of image quality such as Peak Signal-noise Ratio (PSNR), Structural similarity Index Measure (SSIM), and visual comparison. The results show that a fair compromise is provided by the combination of the deep learning and optimization methods to reconstruct the images of high quality when the memory constraint settings are considered.

1.1 Background

Image super-resolution (SR) is a basic issue of computer vision and image processing, which aims at improving the space resolving of pictures. Images obtained in most real-life situations also are taken with constraints of low-quality sensors, compression, or limited bandwidth which lead to low-resolution (LR) images. Such images are not always fine-detailed, sharp or textured to be analyzed and interpreted. Through the use of the interpolation-based approaches to super-resolution (e.g., nearest neighbor, bilinear and bicubic interpolation), are both computationally cheap and simple, they do not recover high-frequency details. Consequently the images that are rebuilt are blurred and not visual. As deep learning evolves, the Convolutional Neural Networks (CNNs) have demonstrated great improvements in the image super-resolution work. SRCNN, FSRCNN, and ESPCN are the models which learn the complicated mappings between LR and high-resolution (HR) images and allow the more detailed reconstruction of the details and textures. Nonetheless, these models can be expensive to run in terms

of computation and memory and therefore cannot be practically applied on resource-constrained machines. In order to overcome these obstacles, there is recent research on production of lightweight and efficient models. Also, to optimize model results and enhance the quality of images without raising the cost of computation considerably, optimization methods, including Automatic Differentiation (AD) and Conjugate Gradient (CG) are under consideration. Thus, the integration of deep learning and optimization techniques can be viewed as a promising step towards obtaining high-quality image super-resolution at no more than memory and calculation cost.

1.2 Motivation

Over the past years, there has been a tremendous demand of high-quality images in diverse applications like in the medical imaging, surveillance, satellite imaging, and multimedia applications. All real-world systems, however, tend to be limited by hardware, bandwidth, and storage capacity, which often leads to low-resolution images. Even though super-resolution models that are based on deep learning have shown remarkable performance, they are generally memory and power-consuming. This renders them inappropriate to run on resource limited devices like mobile, embedded and edge devices. The other significant weakness is that most of the existing techniques are based on enhancing accuracy and do not pay much attention to efficiency and optimization. Consequently, there is a demand of methods that can compromise between image quality and computation and memory efficiency. This inspired the creation of the developed work, which combines a light-weight deep learning program (FSRCNN) with optimization methods Automatic Differentiation (AD) and Conjugate Gradient (CG). The aim will be to improve the quality of images at less memory requirement and lower computing costs, where the solution will be more realistic in real-world scenarios.

1.3 Problem Overview

Image Super-Resolution (ISR) is intended to restore a high-resolution (HR) image using a low-resolution (LR) image. Nonetheless, it is a bad posed problem as several high-resolution images may be associated with the same low-resolution image. It is difficult to retrieve lost information like edges, textures and fine structures. Despite the fact that deep learning models have been effective in enhancing the performance of super-resolution, they tend to be computationally intensive, and they also involve huge memory. This poses a challenge to the implementation of these models in the real-world setting that has scarce resources. Consequently, the main issue that will be discussed in this work is to create the system capable of producing high-quality images of low-resolution input and keeping the use of memory and computational efficiency low.

1.3.1 Objectives

The key aims of this work consist in:

- To come up with an effective image super-resolution model based on deep learning.
- To compare various models including SRCNN, FSRCNN, ESPCN and UNet.
- To enhance the quality of images with the help of optimization methods such as Automatic Differentiation (AD) and Conjugate Gradient (CG).
- To measure the performance by PSNR and SSIM measures.
- To have a trade-off between quality of images and computation.

1.4 Scope of the Work

This paper is dedicated to the topic of single-image super-resolution with the help of deep learning and optimization algorithms. The work is restricted to image quality

improvement with the help of a small-weighted model and does not take into account the video super-resolution or real-time streaming applications. The input data is made up of high resolution images, which are used to produce low resolution images used in training and testing. The work primarily focuses on enhancing the quality of visuals, eliminating noise, and preserving structural similarity at a minimal consumption of memory. The suggested solution can be used in spheres like medical imaging, surveillance systems, and satellite imaging, where the quality of images is significant yet may have little or no computation resources.

Chapter 2

Literature Review

The present paper is based on the Memory-Aware Automatic Differentiation (AD) of Optimization-Based Image Super-Resolution, which was motivated by the recent development in computational imaging and optimization methods. Watson et al. (2025) point out that although the application of exact Hessians and variable projection methods has been made easier with Automatic Differentiation, it creates a considerable amount of memory and computational overhead. Thus, it is necessary to formulate memory-efficient approaches. In the recent past, image reconstruction and optimization efforts have actively utilized the Automatic Differentiation (AD) to make gradient computation more amenable and enhance performance on more complex systems. These methods are however usually computationally and memory expensive and this is why memory conscious optimization methods are sought after.

Christianson (2018) Christianson (2018) explored differentiation through the Conjugate Gradient (CG) method and demonstrated that accurate gradients can be obtained even when the solver is inexact. This is particularly useful in optimization-based image reconstruction, where iterative solvers are commonly used. The study highlights the feasibility of integrating CG with gradient-based frameworks, forming the basis for combining AD and CG in super-resolution tasks. Margossian (2019) Margossian (2019)

provided a comprehensive review of Automatic Differentiation, discussing its implementation and efficiency. The study emphasizes that AD enables exact gradient computation without symbolic derivation, making it highly suitable for large-scale optimization problems. However, it also points out the challenges related to memory consumption, especially in deep computational graphs. Chen et al. (2018) introduced Neural Ordinary Differential Equations (Neural ODEs), which reformulate deep learning models as continuous transformations. Their work demonstrates how memory-efficient backpropagation can be achieved through adjoint methods, reducing memory requirements during training. This concept is closely related to memory-aware AD techniques used in this project. Herring et al. (2018) proposed the Linearize and Project (LAP) method for solving inverse problems with coupled variables. Their approach separates variables and simplifies optimization, which aligns with the concept of variable projection used in super-resolution. However, the method requires efficient differentiation through projection steps, highlighting the importance of AD. Watson et al. (2025) investigated the application of Automatic Differentiation in image registration tasks. Their work demonstrates that AD simplifies optimization by enabling direct computation of derivatives in complex imaging systems. However, they also emphasize the trade-off between accuracy and memory usage, motivating the need for memory-aware approaches. Sun et al. (2024) proposed a neural field-based approach for medical image registration, leveraging deep learning and optimization techniques. Their method demonstrates improved accuracy but requires significant computational resources, highlighting the importance of lightweight and efficient models. Wu et al. (2022) introduced NODEO, a Neural Ordinary Differential Equation-based optimization framework for deformable image registration. This approach combines deep learning with optimization, similar to the proposed work. However, it also suffers from high computational complexity, reinforcing the need for efficient alternatives. Paszke et al. (2019) developed PyTorch, a high-performance deep learning framework that supports Automatic Differentiation through dynamic computational graphs. PyTorch enables efficient implementation of AD-based optimization methods, making it widely used in image reconstruction and

super-resolution tasks. Guzzi et al. (2023) Guzzi et al. (2023) explored the application of Automatic Differentiation in inverse problems for X-ray imaging and microscopy. Their work demonstrates that AD can significantly improve reconstruction accuracy, but it requires careful handling of computational resources, particularly memory. Park (2023) Park (2023) proposed efficient automatic subdifferentiation techniques for programs with linear branches. This work improves the efficiency of gradient computation in complex systems and ensures scalability, which is important for large-scale image processing tasks.

Based on the above analyses, it is clear that Automatic Differentiation is an essential point of activity in the image reconstruction approaches of modern optimization. Yet the majority of the current methods pay attention to accuracy and do not take into account memory efficiency. Also, iterative solvers like Conjugate Gradient can enhance optimization performance, but their combination with AD creates more problems with computations. Hence, there is a gap in the research to come up with memory efficient frameworks that will integrate both deep learning and optimization approaches. The given work fills this gap and combines a lightweight FSRCNN model with Automatic Differentiation and Conjugate Gradient algorithms to generate high-quality and efficient image super-resolution. Based on the literature, it is clear that although Automatic Differentiation offers effective tools to find optimal solutions in image reconstruction, it presents serious memory problems. Available studies are concerned with minimizing the use of memory by use of checkpointing, partial differentiation, and effective optimization strategies. Nevertheless, the question of the integration of lightweight-deep learning models and memory-conscious optimization approaches remains. This gap is filled in the proposed work as it incorporates FSRCNN with AD and Conjugate Gradient to produce efficient and quality image super-resolution.

2.1 Mathematical Formulation

This section presents the key mathematical formulations used in the proposed image super-resolution framework.

2.1.1 Normalization

$$x_{norm} = \frac{x}{255} \quad (2.1)$$

Explanation: This formula normalizes pixel values from the range [0, 255] to [0, 1], which helps in faster and stable model training.

2.1.2 Mean Squared Error (MSE)

$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [I(i, j) - K(i, j)]^2 \quad (2.2)$$

Explanation: MSE measures the average squared difference between the predicted image and the ground truth image. Lower MSE means better image reconstruction.

2.1.3 Peak Signal-to-Noise Ratio (PSNR)

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (2.3)$$

Explanation: PSNR evaluates image quality. Higher PSNR indicates better reconstruction and less error between original and output images.

2.1.4 Structural Similarity Index (SSIM)

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (2.4)$$

Explanation: SSIM measures similarity between two images based on structure, brightness, and contrast. Values closer to 1 indicate better similarity.

2.1.5 Loss Function (Training)

$$L = \frac{1}{N} \sum (y_{pred} - y_{true})^2 \quad (2.5)$$

Explanation: This is the loss function used to train the model. It calculates how far the predicted image is from the actual image.

2.1.6 Gradient Descent Update

$$\theta = \theta - \eta \nabla L \quad (2.6)$$

Explanation: This formula updates the model parameters in the direction that reduces the loss. η is the learning rate.

2.1.7 Total Variation (TV) Regularization

$$TV(x) = \sum |x_{i,j} - x_{i+1,j}| + |x_{i,j} - x_{i,j+1}| \quad (2.7)$$

Explanation: TV helps in reducing noise and preserving edges by minimizing intensity differences between neighboring pixels.

2.1.8 Automatic Differentiation Optimization

$$L = \|x - x_0\|^2 + \lambda \cdot TV(x) \quad (2.8)$$

$$x_{t+1} = x_t - \alpha \nabla L \quad (2.9)$$

Explanation: AD computes gradients automatically and refines the image by minimizing loss. It improves image smoothness and removes noise.

2.1.9 Conjugate Gradient Method

$$x_{k+1} = x_k + \alpha_k d_k \quad (2.10)$$

$$d_k = -g_k + \beta_k d_{k-1} \quad (2.11)$$

$$\beta_k = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}} \quad (2.12)$$

Explanation: CG is an optimization method that improves convergence speed by using previous gradient information. It helps in achieving sharper and more accurate results.

2.1.10 Final Pipeline Representation

$$x_{final} = CG(AD(FSRCNN(LR))) \quad (2.13)$$

Explanation: This represents the complete workflow where the low-resolution image is first enhanced using FSRCNN, then refined using AD and CG to produce the final high-quality image.

2.2 Deep Learning Models for Super-Resolution

Deep learning has contributed enormously to the performance of image super-resolution as complex mappings are learnt between low-resolution (LR) and high-resolution (HR) images. Convolutional Neural Networks (CNNs) are popular because they can recognize the spatial characteristics and reproduce minute characteristics of images. SRCNN is among the first deep learning models that were suggested to super-resolution. It does end to end mapping with a basic three layer convolutional architecture. Even though it helps in enhancing the quality of the images as compared to the traditional methods, it is computationally complex because of the pre-processing measures. FSRCNN is a better variant of SRCNN created to make the algorithm less complex in calculations and faster. It has a more efficient feature extraction and upscaling, as it uses fewer filters and fewer deconvolution layers. SRCNN can be used in real-time and other applications that have memory restrictions, since it is lightweight. ESPN provides a sub-pixel convolution layer that upscales at the final network. This makes the extraction of features easier in the computation aspect and it is more efficient with good quality of images. UNet is a skip-connected encoder-decoder architecture and serves the purpose of preserving spatial information. It introduces both the low and high-level features hence leads to high-quality reconstruction. It is however memory-intensive as it has more parameters. Despite the fact that these models have made tremendous achievements on image quality, most of them have high computational requirements. Consequently, a lightweight and efficient model should be chosen in order to have a practical use.

2.3 Optimization Techniques in Image Processing

Machine learning, optimization methods are important in enhancing image quality in the reconstruction process. These techniques can be employed in image super-resolution where the output of the deep learning images is refined and the quality of the image is improved. Automatic Differentiation (AD) is an efficient method of calculating the gradients of functions. In contrast to numerical differentiation, AD gives the exact gradients through the systematic use of the chain rule. The output image in this work is refined with AD and reduces a specified loss parameter resulting in better smoothness and less noise.

The optimization objective can be expressed as:

$$L = \|x - x_0\|^2 + \lambda \cdot TV(x) \quad (2.14)$$

where x_0 is the initial output from the deep learning model and $TV(x)$ is the total variation term that helps in preserving edges and reducing noise.

Conjugate Gradient (CG) is an iterative optimization technique that is applied to solve large-scale problems. It enhances the speed of convergence with the application of conjugate directions rather than the simple gradient descent. CG can be used on top of AD in order to refine the image even more and make the image sharper.

The update rule in CG is given by:

$$x_{k+1} = x_k + \alpha_k d_k \quad (2.15)$$

where d_k is the search direction and α_k is the step size.

According to the discussion of the available image super-resolution algorithms, a number of main challenges have been outlined. Majority of the deep learning models are concerned with enhancing the quality of images but do not take into consideration the computation efficiency and memory consumption. This has made such models hard to use in practice, particularly on low-resource devices. The other critical problem is that reconstructed images usually become flawed either by loss of fine details, over-smoothing or noise artifacts. Even though the advanced architectures are more

effective in enhancing performance, they demand more complex networks and bigger parameters. As such, the key issue determined in this work is to develop an image super-resolution system, which is capable of delivering effective increase in image quality with a low memory usage and low cost of computation. Also, refinement methods should be enhanced to achieve more detailed results that are sharper.

Chapter 3

Research Problem

3.1 Formal Problem Definition

Image Super-Resolution (SR) is defined as the process of reconstructing a high-resolution (HR) image from a given low-resolution (LR) image. The goal is to recover lost details such as edges, textures, and fine structures while maintaining visual consistency.

Mathematically, the problem can be formulated as:

$$x_{HR} = f(x_{LR}) \quad (3.1)$$

where x_{LR} is the input low-resolution image, x_{HR} is the reconstructed high-resolution image, and $f(\cdot)$ represents the mapping function learned by the model.

The objective of the model is to minimize the difference between the predicted output and the ground truth image. This is achieved using a loss function such as Mean Squared Error (MSE):

$$L = \frac{1}{N} \sum (y_{pred} - y_{true})^2 \quad (3.2)$$

To further improve the output, optimization techniques are applied. The refinement process is guided by the following objective function:

$$L = \|x - x_0\|^2 + \lambda \cdot TV(x) \quad (3.3)$$

where x_0 is the initial output from the FSRCNN model and $TV(x)$ is the total variation term used for smoothness and noise reduction.

The final output is obtained by integrating deep learning and optimization techniques:

$$x_{final} = CG(AD(FSRCNN(x_{LR}))) \quad (3.4)$$

Thus, the problem focuses on generating high-quality images from low-resolution inputs while minimizing memory usage and computational complexity.

3.2 Limitations of Existing Methods

Although significant progress has been made in image super-resolution using deep learning, several limitations still exist in current approaches.

High Memory Consumption: Most deep learning models such as VDSR, RCAN, and UNet require large memory due to deep architectures and high number of parameters. This limits their applicability in resource-constrained environments.

High Computational Complexity: Advanced models often involve complex operations and deep layers, leading to increased training and inference time. This makes real-time deployment challenging.

Over-Smoothing of Images: Many models tend to produce overly smooth images, which results in loss of fine details and textures, especially in high-frequency regions.

Limited Generalization: Models trained on specific datasets may not perform well on unseen or real-world degraded images, affecting robustness.

Inefficient Optimization: Traditional methods rely solely on deep learning without incorporating optimization techniques, leading to suboptimal refinement of output images.

Lack of Memory-Aware Approaches: Most existing works focus on improving accuracy while ignoring memory efficiency. This creates a gap in developing lightweight and efficient models.

To address these limitations, the proposed work introduces a hybrid approach that combines a lightweight deep learning model (FSRCNN) with optimization techniques such as Automatic Differentiation (AD) and Conjugate Gradient (CG), aiming to achieve both high-quality reconstruction and memory efficiency.

3.3 Problem Identification

According to the discussion of the available image super-resolution algorithms, a number of main challenges have been outlined. Majority of the deep learning models are concerned with enhancing the quality of images but do not take into consideration the computation efficiency and memory consumption. This has made such models hard to use in practice, particularly on low-resource devices. The other critical problem is that reconstructed images usually become flawed either by loss of fine details, over-smoothing or noise artifacts. Even though the advanced architectures are more effective in enhancing performance, they demand more complex networks and bigger parameters. As such, the key issue determined in this work is to develop an image super-resolution system, which is capable of delivering effective increase in image quality with a low memory usage and low cost of computation. Also, refinement methods should be enhanced to achieve more detailed results that are sharper.

3.4 Research Gap

Based on the literature review, it is seen that great progress has been achieved in image super-resolution by deep learning models. There is, however, most of these methods that are focused on accuracy in lieu of efficiency, which results in excessive memory usage and computation. Moreover, lightweight deep learning models are not commonly combined with optimization methods that operate on Automatic Differentiation (AD) and Conjugate Gradient (CG). The current approaches essentially apply only deep learning or optimization, yet, there is rarely an effective combination of both. Approaches that seek to strike a balance between image quality, computation efficiency

as well as memory consumption are lacking as well. This leaves a research gap in the development of hybrid models that are capable of producing high-quality reconstruction and at the same time be implemented in the real world. The proposed work is inspired by this gap, where they combine a lightweight deep learning model along with optimization techniques to enhance performance and efficiency.

3.5 Proposed Solution

In order to solve the problems outlined, a memory efficient image super-resolution architecture is presented. The strategy is a lightweight deep learning model that is optimized to improve performance and efficiency. The first stage involves the FSRCNN model processing the low-resolution image to produce a high-resolution one. The reason behind the choice of FSRCNN is that the model has a faster processing speed and less memory requirements than other deep learning models. Automatic Differentiation (AD) is used in order to further increase the quality of output. AD is used to compute gradients efficiently and refines the image with a defined loss function leading to a reduction in noise and an increase in structural consistency. This is followed by the Conjugate Gradient (CG) approach, which effectively improves convergence and details of the images. CG uses the past gradient data that is used to enhance optimization efficiency and provide better output.

The overall product is the result of the combination of stages:

$$x_{final} = CG(AD(FSRCNN(x_{LR}))) \quad (3.5)$$

The solution suggested is intended to obtain a high-quality image reconstruction without compromising the low computational and memory demands of the solution, which is why it can be used in practice.

3.6 Basic Definitions

Image Super-Resolution

Image Super-Resolution (SR) is the process of creating a high-resolution (HR) image with low-resolution (LR) as an input image. It is used with the aim of enhancing the image by boosting the image clarity and recovering the lost high-frequency details in the image degradation.

Low-Resolution and High-Resolution Images

The pixel count in the low-resolution image is lower and details are less detailed and are normally blurred or unclear. High-resolution image contains a higher number of pixels and finer details that form its image of high quality and clarity.

Deep Learning

Deep Learning is a branch of machine learning, which uses a neural network with multiple layers to be trained on complex patterns of data. Deep learning CNNs are used in the image processing task to learn high-low mappings of images.

FSRCNN

It is a lightweight model, Fast Super-Resolution Convolutional Neural Network (FSRCNN), which is an image super-resolution convolutional neural network. It is also computationally easy and has good performance and hence can be applied in real time applications.

Automatic Differentiation (AD)

Automatic Differentiation is a computing methodology that is used to compute gradients of functions with respect to their inputs automatically. In this project, the AD is used to optimize the generated image by means of gradient-based changes.

Conjugate Gradient (CG)

Conjugate Gradient is an optimization algorithm that can be used in the solution of big problems. It is fast to converge compared to the conventional gradient descent and it helps in refining the output image in the proposed system.

Peak Signal-to-Noise Ratio (PSNR)

PSNR is a parameter that is used in the assessment of the quality of reconstructed and compared image against an original image. It is based on the Mean-Squared-Error (MSE) of two pictures. As the values of PSNR increase, there is an implication of quality of the image.

Structural Similarity Index Measure (SSIM)

SSIM is a number which is utilized to calculate the similarity between two pictures in respect to structural image, brightness, and contrast. The values of SSIM range between 0 and 1 with the increased the values the higher the similarity.

Chapter 4

Methodology

This section is the description of a proposed workflow of image super-resolution that is memory efficient. The algorithm is a mix of a sparse deep learning model and optimisation solutions to improve the image quality without rendering the algorithm computationally intensive. Data preprocessing is the critical aspect of the proposed super-resolution model because it eliminates any unclean, inconsistent or inappropriate data as an input into the model.

4.1 Data Preprocessing

The dataset referred to as DIV2K in this study is the one that contains high-quality images that are widely used in the image super-resolution operations. The dataset includes high-resolution (HR) images that are used as ground truth. Since the model is directed at training the mapping between low-resolution (LR) and high-resolution images, LR images are created artificially by bicubic interpolation of HR images. This is done to model real degradation in the real world whereby images are downsampled or compressed, and once LR images are produced, a series of preprocessing steps are done in both LR and HR images simply to demonstrate consistency:

Resizing: The size of the images is made the same, e.g. 256 x 256 pixels. This standardization helps in making sure that the size of the input of the neural network is identical and simplifies the computation process.

Normalization: The conversion of pixel values of images to the range [0,1] is done by:

$$x_{norm} = \frac{x}{255} \quad (4.1)$$

where x represents the original pixel intensity. Normalization enhances stability of numbers in training and also increases convergence.

Pairing of Data: All the low-resolution images are matched with the high-resolution image. This combination is essential to supervised learning because the model would be trained to repeat HR images using LR inputs.

Data Formatting: The images are later converted into some sort of tensors to be inserted in the structure of profound learning technologies such as PyTorch. The channel order is modified and information is batched so that it could be trained effectively.

Dataset Splitting: The data set will be divided into training and validation set. Training parameters of the learning model occurs on the training set and performance is measured on the validation set.

All these preprocessing operations are required to ensure that the data is organized, normalized and aligned and it can be learned by the model to produce high quality and super-resolution images.

4.2 Model Comparison and Model Selection

This paper will compare and contrast different deep learning models to establish which model to adopt in image super-resolution without affecting the memory and computer processing speed. The considered architectures are the SRCNN, FSRCNN, ESPCN, and UNet-based ones.

SRCNN is one of the earliest convolutional neural network (CNN) models applied to perform super-resolution. It learns a mapping between low-resolution and high-resolution images but it is computationally expensive to execute because it is designed as such.

ESPCN improves the efficiency through the ability to offer sub-pixel convolution to achieve quicker upscaling.

UNet Encoder-decoder based models are the ones that lead to high quality reconstruction at the cost of high computational cost and also consumes memory.

FSRCNN Fast Super-Resolution Convolutional Neural Network has been developed in order to address these deficiencies. It offers architectural improvements such as reduction and expansion of layers which reduce the number of parameters but has no impact on the performance. SRCNN therefore is computationally cheap and consumes less memory than their counterparts do.

Model	Initial Loss	Final Loss
SRCNN	0.0406	0.0015
FSRCNN	0.0948	0.0147
ESPCN	0.0841	0.0085
UNet-SR	0.0482	0.0005

Table 4.1: Model Comparison based on Training Loss

The table presents the comparison of various models depending on the training loss. UNet-SR attained the lowest final loss, which means it has a greater learning potential. SRCNN + offers a tradeoff between performance and computation efficiency.

Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) are the typical measures of evaluation that one can use to compare these models. PSNR is defined as:

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (4.2)$$

where MAX is the maximum pixel value (typically 255), and MSE is the Mean Squared Error given by:

$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [I(i, j) - K(i, j)]^2 \quad (4.3)$$

where $I(i, j)$ represents the original image and $K(i, j)$ represents the reconstructed image.

SSIM is employed together with PSNR in order to determine similarities in images on the structural level. SSIM considers luminance, contrast and structure as one perceptual measure of quality of an image.

Model	PSNR (dB)	SSIM	Size (MB)
SRCNN	28.92	0.940	0.26
FSRCNN	19.05	0.516	0.07
ESPCN	21.92	0.683	0.10
UNet	33.65	0.978	0.58

Table 4.2: Model Comparison based on PSNR, SSIM and Model Size

The table compares the various super-resolution models in terms of PSNR, SSIM and model size. The UNet shows the best PSNR (33.65 dB) and SSIM (0.978), which means that it is more correlated and has the best image quality. It is however the one with the biggest model size (0.58 MB), thus it is not efficient as much as regards memory. SRCNN is also suitable in the cases of high PSNR and SSIM values but is bigger than FSRCNN. ESPCN offers moderate performance of quality and size balance. FSRCNN does not have the highest PSNR and SSIM with other models, but the model size (0.07 MB) is the lowest, turning it into a very efficient and memory-friendly application. Thus, it is possible to choose FSRCNN as the base model in the given work because this method is light and computationally efficient.

The models are compared based on the following criteria:

- Reconstruction quality (PSNR and SSIM)
- Computational efficiency
- Model size and memory requirements

According to the comparative analysis, it is possible to mention that UNet is more precise, but consumes a significant amount of memory. SRCNN and ESPCN have

medium performance and are not efficient as FSRCNN. FSRCNN is good performance-wise and efficient. It is competitive in terms of PSNR and SSIM and costly in terms of its computational and memory cost. Therefore, FSRCNN is chosen as a base model in this paper. The developed model is also optimised through the assistance of Automatic Differentiation and Conjugate Gradient as a technique of enhancing the output quality without complicating the model.

4.3 Model Training

After making a decision to utilize FSRCNN as the baseline model, one should be capable of training the model on the base of paired low-resolution (LR) and high-resolution (HR) images. The objective of the training process would be to train an efficient mapping between LR images and their associated HR images. The training process would be carried out in a supervised manner in such a way that each and every LR image would be taken as input and the corresponding HR image would be taken as the target output. The model is also trained by minimization of the difference between the actual image and the predicted one.

Loss Function: The reconstruction error of the predicted image and the ground truth image is measured with the help of the Mean Squared Error (MSE) loss function:

$$L = \frac{1}{N} \sum_{i=1}^N (y_{pred}^i - y_{true}^i)^2 \quad (4.4)$$

where y_{pred} is the predicted image, y_{true} is the ground truth image, and N is the total number of pixels. The goal of training is to minimize this loss function.

Training Configuration:

The model is trained using the following configuration:

- Number of epochs: 30
- Batch size: 8
- Learning rate: 0.001

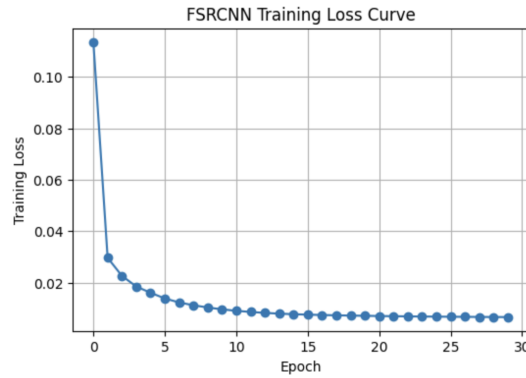


Figure 4.1: FSRCNN Training Loss Curve over 30 Epochs

Figure 1 Shows The training loss curve illustrates the learning process of the model in terms of time-based process. The initial forecasts indicate massive losses that indicate that the model is incapable of making precise forecasts. The progress of the model is demonstrated by the training when the model attains a rapid decline of the loss in the first training period. The model has attained the final level of performance due to the fact that the loss that comes after several training sessions has stabilized. The model works best as per the final low loss value that means that it is ready to undergo further testing.

Training is performed in mini-batches to improve computational efficiency and stability.

Training Process:

During each epoch, the following steps are performed:

1. A batch of LR images is passed through the FSRCNN model.
2. The model generates predicted HR images.
3. The loss between predicted and actual HR images is computed.
4. Gradients are calculated using backpropagation.
5. Model parameters are updated using gradient descent.

Convergence:

The loss value decreases as training continues and it implies that the model is learning to predict LR by the HR images. The loss is held constant during the continuation of the training because the trained FSRCNN model is optimally used and then optimization methods applied on the top of the trained model, which include Automatic Differentiation and Conjugate Gradient.

4.4 Applying Automatic Differentiation

The attainment of the FSRCNN model first level super-resolved image requires the next refinement step to enhance the image quality, noise reduction and structural integrity. In this direction, Automatic Differentiation (AD) will be used as an algorithm of refinement that is grounded on optimization. Automatic Differentiation Automatic Differentiation is a calculation method that computes the derivatives of a specified function in regard to its inputs as effectively as it can. Unlike numerical differentiation, which is inaccurate (becoming so because of approximations), and symbolic differentiation, which can be computationally time-intensive, AD is a precise and fast way of calculating gradients.

Role of Automatic Differentiation:

Gradients can be calculated using AD and not writing down the complex equations by hand. This makes the refinement efficient with respect to both the computational costs and high dimensionality image data.

Effect on Image Quality:

Applying AD results in:

- Reduction of noise and artifacts,
- Improved structural consistency,
- Enhanced smoothness while preserving edges.

Automatic Differentiation is, hence, significant in terms of improving the quality of the super-resolved image without having a great deal of complexity in the model.

4.5 Applying Conjugate Gradient

After automatic differentiation (AD) refining of the picture, a second refining stage is then followed by Conjugate Gradient (CG). The step will also maximize convergence, sharpen images and reduce any artifacts that may still be present. The Conjugate Gradient algorithm is a very useful iterative optimization method, which is usually used to solve large scale optimization problems. CG also considers the information across the previous iterations to make more constructive search directions than using standard gradient descent which only considers the existing gradient to update the solution, thereby converging faster. **Motivation for Using CG:**

AD despite the improvement in the image quality may not yield faster convergence speed or slight blurring of detail. CG can address this weakness by:

- Accelerating convergence,
- Reducing the number of iterations,
- Preserving important image structures.

Advantages of CG:

- Faster convergence compared to gradient descent,
- Efficient for high-dimensional data such as images,
- Utilizes previous optimization steps for improved performance,
- Reduces oscillations during optimization.

Effect on Image Quality:

Applying CG after AD refinement results in:

- Sharper edges,
- Improved texture details,
- Reduced over-smoothing,

- Better overall visual quality.

Conjugate Gradient method, thus, perfects the process where the image is optimized better and is of superior quality output without the necessity of developing and optimization of the algorithm at a high level of complexity.

4.6 Generating Output Image

The final stage of the proposed development is developing the picture of the high-resolution output after all the refinements have been applied. This is a combination of the results obtained by the deep learning model and optimization processes to form a better image in terms of visual quality, in which the low-resolution (LR) image is passed through the FSRCNN model to obtain a preliminary super-resolved image. Probably this output may still be of poor quality with noise and other finer details being overlooked. Automatic differentiation (AD) is used to optimize the image by reducing the objective-function being optimized and this improves structural consistency and reduces artifacts. Then, Conjugate Gradient (CG) process optimization is used to optimize the optimized image. The sharpness and convergence of CG is improved by incorporation of past gradient information that results in the maintenance of edges and negligence of the smoothing effects.

The generated image is compared to Structural Similarity Index Measure (SSIM) that is obtained as:

$$x_{final} = CG(AD(FSRCNN(LR))) \quad (4.5)$$

where:

- LR is the input low-resolution image,
- $FSRCNN(\cdot)$ represents the deep learning model output,
- $AD(\cdot)$ represents the refinement using Automatic Differentiation,
- $CG(\cdot)$ represents the optimization using Conjugate Gradient.

Post-processing:

After refinement, the output image may undergo minor post-processing steps such as:

- Clipping pixel values to the valid range,
- Converting normalized values back to the original scale:

$$x_{final} = x_{final} \times 255 \quad (4.6)$$

Quality Evaluation:

The image produced is compared with Structural Similarity Index Measure (SSIM) which is calculated as:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4.7)$$

where x is the generated image and y is the ground truth image.

Output Characteristics:

The image of the final output of the proposed method contains:

- Greater transparency and decision making,
- Reduced noise and artifacts,
- Improved edge sharpness,

In such a way, a combination of deep learning and optimization methods makes sure that the resulting image obtained is of a visual quality and is computationally efficient.

4.7 Workflow Architecture

The workflow offered is a series of steps that will improve the quality of the image in a productive way. First, a low-resolution (LR) image is received and preprocessed by resizing and normalization. The processed image is then inputted into the FSRCNN model which produces an original high-resolution (HR) image. The reconstruction is

the base reconstruction. Automatic Differentiation (AD) is used to further enhance the quality. AD calculates loss function gradients and optimizes the image through the reduction of reconstruction error and noise. Once AD refinement is completed, Conjugate Gradient (CG) method is applied in order to speed up convergence and improve the sharpness of the image by taking into account past gradient data. Lastly, post-processing measures are applied to the processed image to improve the image e.g. sharpening and contrasting to get the final output.

The overall pipeline can be represented as:

Low Resolution Input → Data Preprocessing → FSRCNN → AD Refinement → CG Optimization → Final Output Image

4.8 Evaluation Metrics

The quality of the proposed image super-resolution model is measured on the basis of the typical quantitative values like Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). PSNR is used to measure the quality of the reconstructed image based on the ground truth image. It is defined as:

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (4.8)$$

The greater the PSNR values, the greater the quality of reconstruction. SSIM compares the two images, the predicted and the ground truth image, on the basis of luminance, contrast and structure:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4.9)$$

The values of SSIM are between 1 and 0, and values that are more similar are those that are nearer to 1.

These measures give both quantitative and subjective measurements of picture quality.

4.9 Implementation Details

The proposed model is written in Python and PyTorch deep learning framework. The experiments are done in Kaggle notebooks that have the support of the GPU to speed up the training and inference process. The dataset is DIV2K that has images of high resolutions. Bicubic interpolation is used to create low-resolution images. All the images are resized to a constant square size of 256 x 256 and they are object normalized to figure [0,1]. Mean Squared Error (MSE) is used as a loss function to train the FSRCNN model. The gradient descent optimization is used to update the model parameters. The process of training is repeated with several epochs until the loss stabilizes. The model output is then optimized over Automatic Differentiation (AD) and Conjugate Gradient (CG) techniques, after training. There are other methods of post-processing like sharpening and contrast manipulation that are used to enhance visual quality.

4.10 Advantages of Proposed Method

The suggested approach has a number of benefits in comparison with the traditional and existing deep learning methods:

- **Memory Efficiency:** The FSRCNN makes it possible to utilize low memory consumption as opposed to the deeper models.
- **Improved Image Quality:** AD and CG combined together increase the clarity of the images, their sharpness, and structure.
- **Reduced Noise:** The optimization methods contribute to the reduction of noise and artifacts of the rebuilt image.
- **Faster Computation:** The light architecture provides expediency in processing and real time applicability.
- **Better Balance:** The approach has a record of balancing both performance and computational efficiency.

Chapter 5

Results and Discussions

In this sub-section, the results of the proposed memory efficient image super-resolution model are provided. The FSRCNN model selected is also evaluated in the aspects of performance and refinement schemes that relying on Automatic Differentiation (AD) and Conjugate Gradient (CG) and it is quantified with a quantitative measure and the qualitative evaluation in terms of visual analysis.

Quantitative Evaluation:

In order to evaluate the performance of the model, Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) are employed. The measures help to establish if the reconstruction of the generated images is high when compared to the ground truth images.

The following are the results obtained on the different stages of the proposed pipeline:

Method	PSNR	SSIM
FSRCNN	19.16	0.655
FSRCNN + AD	20.22	0.760
FSRCNN + AD + CG	23.30	0.788

Table 5.1: Performance Comparison

Based on the results given above, one can note that the incorporation of optimization methods goes a long way in enhancing the performance of the model. The value of

PSNR improves, which means that there is a higher quality of reconstruction, and the value of SSIM improves, which means that there is a higher level of structural similarity.

Qualitative Analysis:

By the findings provided above, it is possible to observe that the implementation of optimization techniques is a long way to improving the model performance:

- FSRCNN output is sharper with a possibility of little blurring.
- The AD refinement eliminates noise and increases smoothness.
- The edges are further sharpened and fine details enhanced by the CG optimization.

The value of PSNR increases, and this is to say that there is a better quality of reconstruction, the value of SSIM increases which is to say there is better quality of structures similarity.

Model Efficiency:

The output generated is more elaborate in the clarity, edges and the amount of texture compared to the original output. Another important aspect that is notable in this work is memory efficiency. FSRCNN has very low memory usage requirements than other models such as UNet and performs competitively. The output is optimized by the optimization techniques and does not add complexity to the model thus making the system applicable in the real world context.

Summary of Results:

Overall, the proposed algorithm works well regarding the choice of a trade-off between the quality of the image and computing performance. FSRCNN in combination with AD and CG refinement can achieve drastic quantitative and visual metrics, and consequently, the achieved results in the experiment support the application of the method in memory-efficient image super-resolution.

The suggested framework of image super-resolution produces its final results that are presented in this section. The pipeline generates three various output results that incorporate the input with the low-resolution and the FSRCNN output and the resulting final outcomes generated using the Automatic Differentiation and Conjugate gradient

approaches. The low-resolution input images are not sharp and contain low details. The FSRCNN model allows the user to upscale the images but retains certain level of blurriness on the image. The use of AD improves the structural integrity of the image and the noise is minimized. The CG technique allows one to refine additional but still leads to the better edge sharpness and the preservation of all the texture details. The final output images are much more improved in clarity, sharpness, and the overall image quality, as opposed to the input images. The findings prove that deep learning with optimization methods yield superior image quality outputs without compromising the computational efficiency.



Figure 5.1: Low Resolution Image

The figure above illustrates the input that is of low-resolution nature. The picture seems blurred as it does not have the capability to show distinct images consisting of edges and texture. According to the demonstration, super-resolution methods are required to enhance visual sharpness and the lost details.



Figure 5.2: FSRCNN Output Image

The output of FSRCNN provides a better image of the low-resolution picture. The general structure becomes more obvious and the sharpness of the resolution is improved, however, there is still a degree of blurriness and loss of detail. The present condition needs more efforts in order to attain high sharpness and clarity.

FSRCNN + AD



Figure 5.3: FSRCNN + AD Image

The result of FSRCNN + AD is still better compared to the basic FSRCNN output. The picture indicates that there is an increase in clarity, as it has structural integrity, and it also has reduced noise levels. The blurring that appears in the present condition of the picture needs additional effort to draw more solid lines.



Figure 5.4: FSRCNN + AD + CG Image

The stage where the FSRCNN + AD + CG output has the most improvement is the best amongst all stages. In the picture, there are clearer edges and more detailed texture. The noise is minimized and structural detail is highly preserved thus giving a high quality and visually enhanced end image.

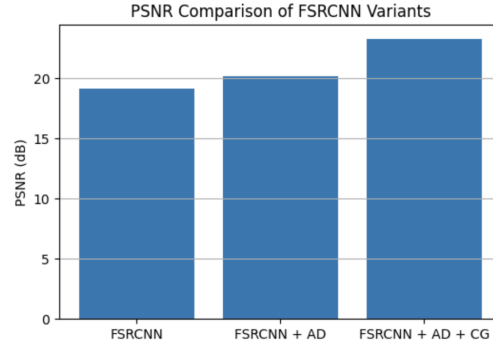


Figure 5.5: PSNR Comparison of FSRCNN Variants

The graph shows the PSNR results of various versions of the FSRCNN model. The original FSRCNN model exhibits the worst PSNR value that leads to reduced quality of the image. With the usage of Automatic Differentiation (AD), the PSNR improvements are also seen that exhibit improved reconstruction outputs. The highest PSNR value is reached with FSRCNN + AD + CG that demonstrates the fact that the combinations of optimization techniques significantly enhance the quality of the image.

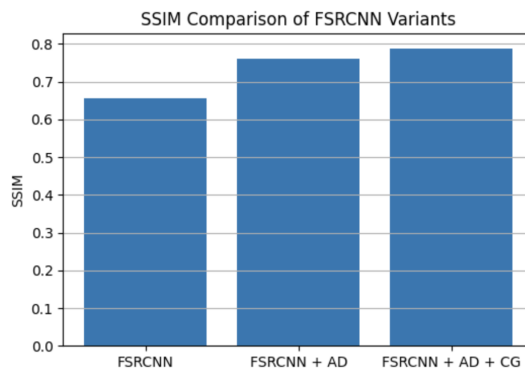


Figure 5.6: SSIM Comparison of FSRCNN Variants

The graph shows the measures of SSIM of different models of FSRCNN. Basic FSRCNN model is the least structurally similar and its structural similarity is enhanced

by the addition of AD. FSRCNN + AD + CG leads to the highest SSIM, which means that image structure and image details are better preserved.

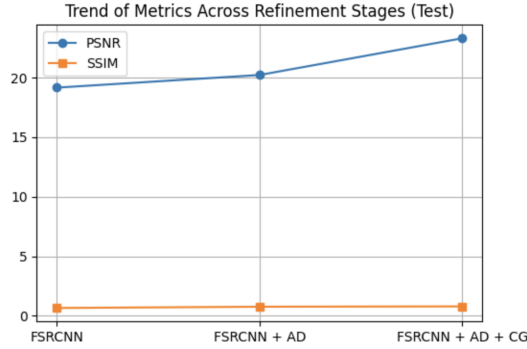


Figure 5.7: Metrics Across Refinement stages

The graph shows the development of PSNR and SSIM measures in different stages of improvement. The two metrics begin with the FSRCNN base level and culminate into optimal performance with FSRCNN + AD + CG combination. The findings confirm that each refinement step increases the quality of the image with the joint approach achieving the best outcome.

The following section will provide a close description of the results in the proposed image super-resolution model. The reason as to why the discussion attempts to understand what improvements were made at each stage of the pipeline and what level of efficiency the proposed approach would have is as follows. It is obvious that the single FSRCNN model can give some portion of improvement to the image resolution process. However, even the generated images may have certain blurring and no fine details. This is largely due to the shortcoming of lightweight models which are casual to efficiency than to learning sophisticated features. To minimize this shortcoming, Automatic Differentiation (AD) is suggested as a refining step. As the AD is added, the structural consistency of the output images is received by minimizing the prescribed loss. It is also effective in noise removal and leaves the image in a smooth manner without depriving the image of beneficial features. It is also mentioned that a process of over smoothing may occasionally result in a loss of sharpness in certain points, the Conjugate Gradient (CG) method further enhances the process. As a result, the final product

images are sharpened in edges, better in texture information and quality of images. The quantitative approach of comparing AD only and AD+CG have revealed that the noise reduction with sharpness maintenance in the images is more balanced in the AD+CG. In the same tone, the higher the values of SSIM, the higher the structural similarity which is required in the perceptual quality. The other useful point is linked to the memory efficiency. These upgrades prove the usefulness of the combination of optimization techniques and deep learning models. Other models like UNet are more precisely calculated, although they require a lot of computation power. Conversely, FSRCNN is a lightweight model and the combination of AD and CG does not increase the size of the model, it only enhances the performance. This makes the approach that is proposed suitable to be implemented in resources constrained environment such as mobile devices. However, the proposed method also has its limitations. Refinement process also introduces certain additional steps in the computations and may lead to the processing time being long. Also, optimization methods depend on parameters selection in order to be efficient e.g. step size, regularization factors, etc. In general, the discussion shows that the proposed framework is able to balance image quality, computational efficiency and memory consumption. The combination of deep learning with optimization methods can be considered a successful approach in the context of pursuing high-quality super-resolution outcomes.

Chapter 6

Summary and Future Work

The objective of the project is to enhance the quality of low-resolution images and to resolve the minimum of the computational and memory constraint of the model. FSRCNN model is selected due to its ability to process images at a high speed and a low memory rate. Despite the fact that FSRCNN provides a certain progress in regards to enhancement of a resolution of images, the final result may still lack finer details. So that this drawback can be overcome, the Automatic Differentiation (AD) is applied to improve the formed image by minimizing a certain objective. The step just helps in the reduction of the noise along with improvement of the structural consistency, further improvement is achieved by the Conjugate Gradient (CG) optimization method, which does not only improve the convergence, but also sharpens the image. Integration of FSRCNN, AD and CG results in a hybrid model that can provide a balance between the performance and the efficiency. Experiment results show that the devised approach can provide a higher value of PSNR and SSIM in comparison to the baseline. It is also confirmed by the visual analysis that the final output images are of higher quality, sharp edges, and texture detail and the complexity of the model is not increased dramatically. Overall, the proposed solution can be described as a suitable solution to the real world projects especially those with limited resources such as mobile devices and embedded systems. The framework proposed proves to be efficient when it comes to addressing the problem of image quality and computational efficiency trade-off.

The proposed approach establishes a good compromise between the quality of images and the speed of calculation but still needs several areas of improvement and further development. One of the possible upgrades is the advanced deep learning architectures. FSRCNN is meant to be used as the lightweight execution but scholars ought to explore the idea of residual networks and attention-based models which can possibly lead to achieving better image output. The optimization methods should undergo upgrades in order to achieve their potential. Automatic Differentiation (AD) and Conjugate Gradient (CG) methods are currently being used in the implementation of the system. Further research may investigate adaptive optimization methods that involve hybrid means of learning rate and regularization strength to improve the work of the system. Data augmentation is another way of improving the preprocessing stage because it involves rotating, flipping, and adding noise to the data. This would be beneficial to the model performance due to the establishment of greater defenses to different image categories and long image processing. The second step is to test the proposed system in actual real time working environments. The system needs pipeline optimization to have faster inference process since it must process video super-resolution and mobile based image enhancement tasks. The second step involves testing the model on various datasets and real pictures which have undergone degradation. There is the need to test the proposed method to ascertain its capability to extrapolate beyond the controlled testing conditions. The study needs to integrate other evaluation measures whereby perceptual image quality measurement would be more accurate by the use of perceptual loss and LPIPS and human visual assessment techniques. The suggested framework can be extended to the adjacent branches that encompass medical image enhancement and satellite image reconstruction and surveillance system development.

References

- Chen, R. T., Rubanova, Y., Bettencourt, J., and Duvenaud, D. K. (2018). Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31.
- Christianson, B. (2018). Differentiating through conjugate gradient. *Optimization Methods and Software*, 33(4–6):988–994.
- Guzzi, F., Gianoncelli, A., Billè, F., Carrato, S., and Kourousias, G. (2023). Automatic differentiation for inverse problems in x-ray imaging and microscopy. *Life*, 13(3):629.
- Margossian, C. C. (2019). A review of automatic differentiation and its efficient implementation. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 9(4):e1305.
- Park, S. (2023). Efficient automatic subdifferentiation for programs with linear branches. *Mathematics*, 11(23):4858.
- Paszke, A., Gross, S., Massa, F., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32.
- Sun, S., Han, K., You, C., et al. (2024). Medical image registration via neural fields. *Medical Image Analysis*, 97:103249.
- Watson, W., Cherry, C., and Lang, R. (2025). Applications of automatic differentiation in image registration. *SIAM Undergraduate Research Online*, 18.

- Wu, Y., Jiahao, T. Z., Wang, J., et al. (2022). Nodeo: A neural ordinary differential equation-based optimization framework for deformable image registration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20804–20813.

Appendix A

Appendix

This chapter provides more information concerning the application of the suggested image super-resolution structure. It contains valuable snippets of code, parameter configuration, and aid tables that are used in the project.

A.1 Model Training Code

The following code snippet shows the training process of the FSRCNN model:

```
for epoch in range(epochs):
    total_loss = 0
    for lr, hr in train_loader:
        lr, hr = lr.to(device), hr.to(device)

        sr = model(lr)
        loss = criterion(sr, hr)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
```

```
total_loss += loss.item()

print(f"Epoch {epoch+1}, Loss: {total_loss}")
```

Explanation:

This code is a training recurrence of the FSRCNN model. Backpropagation is used to compute the gradients and an optimizer is used to update the model parameters. To keep track of the training, the value of the loss is summed up and printed at every epoch.

A.2 Optimization Code (AD and CG)

```
# Automatic Differentiation Refinement
def refine_with_ad(image):
    # gradient-based refinement
    return refined_image

# Conjugate Gradient Refinement
def refine_with_cg(image):
    # CG optimization
    return refined_image
```

Explanation:

This section explains the refinement methods utilized after getting the first output in the model. Gradients are calculated with the help of Automatic Differentiation (AD) and the picture is sharpened with the help of the minimization of the loss function. CG is then used as an optimization technique to increase convergence and sharpen the image. The methods are used in the enhancement of image quality over the base model output.

A.3 Experimental Setup

Explanation:

The following table is a summary of the configuration which is applied in training.

Table A.1: Training Configuration

Parameter	Value
Dataset	DIV2K
Image Size	256×256
Batch Size	8
Epochs	30
Loss Function	MSE
Optimizer	Adam
Learning Rate	0.001

Images of size 256 by 256 are used in the DIV2K dataset. The model would be trained in a batch of 8 and 30 epochs. The loss function is the Mean Squared Error (MSE) and the optimization tool to update model parameters is Adam with learning rate= 0.001. These environments guarantee effective and stable model training.

A.4 Model Comparison Results

Table A.2: Model Performance Comparison

Model	PSNR (dB)	SSIM	Size (MB)
SRCNN	28.92	0.94	0.26
FSRCNN	19.05	0.516	0.07
ESPCN	21.92	0.683	0.10
UNet	33.65	0.978	0.58

Explanation:

This table is a comparison of various models in terms of PSNR, SSIM and model size. UNet is the best in terms of PSNR and SSIM, which means it has a better image quality, but is more large. FSRCNN is less resource-consuming because it is smaller in size, but its performance is worse than that of UNet. The comparison facilitates in choosing a model that is a combination of performance and efficiency.

A.5 Output Samples

The following table summarizes qualitative observations of different stages:

Table A.3: Output Quality Comparison

Stage	Observation
LR Image	Blurry, low details
FSRCNN Result	Better resolution yet blurred a little.
AD Output	Smoother structure, Reduced noise
CG Output	Sharper edges and better details
Final Output	High clarity and enhanced visual quality

Explanation:

The following table shows qualitative notes at various pipeline stages. The image in low-resolution is blurred whereas the FSRCNN output is better in resolution. The AD phase cuts noise, smooths the picture, and the CG one sharpens and increases details. The resulting image has a better visual quality and clarity as compared to the original image.

A.6 Additional Notes

Python and PyTorch are used to implement it. Kaggle Gpu is used to carry out training. The suggested approach is able to optimally trade image quality and computational efficiency.